U M L – D R A F T

UML DIAGRAMS BASED ON SIMPL-OPEN ARCHITECTURE and POC IMPLEMENTATION

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13/06/2025

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1. DOCUMENT PURPOSE

The main purpose of this document is to compile the various UML diagrams related to WP5. The following pages will define the different functions that the Provider can perform, such as ADD and REVOKE operations, interactions with the SIMPL-OPEN dataspace, interactions with EBSI blockchain, and finally the data frame underlying the various data exchanges.

Document Structure

Each chapter will follow this structure:

1. Function Name

1.1 Short Textual Description

1.2 System Operations UML

1.2.1 UML diagram with new interactions coloured or highlighted

1.2.2 UML diagram commentary

1.2.3 Data Exchange (description of all data frames exchanged in the UML flow at a macro level)

1.3 Smart Contract

1.3.1 Smart contract flow (block diagram)

Blockchain data blocks (the actual data saved in the blockchain, referring to subsection 1.2.3 in JSON format)

1.3.2 Smart contract implementation

NOTE:

- The various operations of the business process will particularly refer to UC#1 and therefore to one of the main providers in this use case, namely MASA. MASA represents the operational actor and the data provider in use case #1.
- When referring to blockchain, we refer always to European Blockchain Services Infrastructure (EBSI).

2. FUNCTION NAME: SIMPL-OPEN_ONBOARDING

2.1 Short Description

The documentation of the onboarding process refers to the business process 3a of SIMPL_OPEN, available at the following link:

LINK BP3A - ONBOARDING

The **onboarding process** for a new Applicant details the tasks and decisions required to onboard a new organisation to a Dataspace. Both Providers and Consumers can apply to a Dataspace and will be referred to as (Dataspace) Applicants from here on.

The Applicant Representative submits the onboarding requests to the Governance Authority. Upon approval, they set up the Simpl-Agent, a local gateway enabling interaction with the dataspace. As such installing the Simpl-Agent is a crucial step for onboarding to the dataspace. The Applicant Representative must install and configure the Simpl-Agent. After successful setup, the Applicant receives the necessary security credentials, completing the onboarding process and allowing participation in the dataspace.

The **actors** involved in this process are the:

- Governance Authority,
- Applicant,
- Applicant Representative.

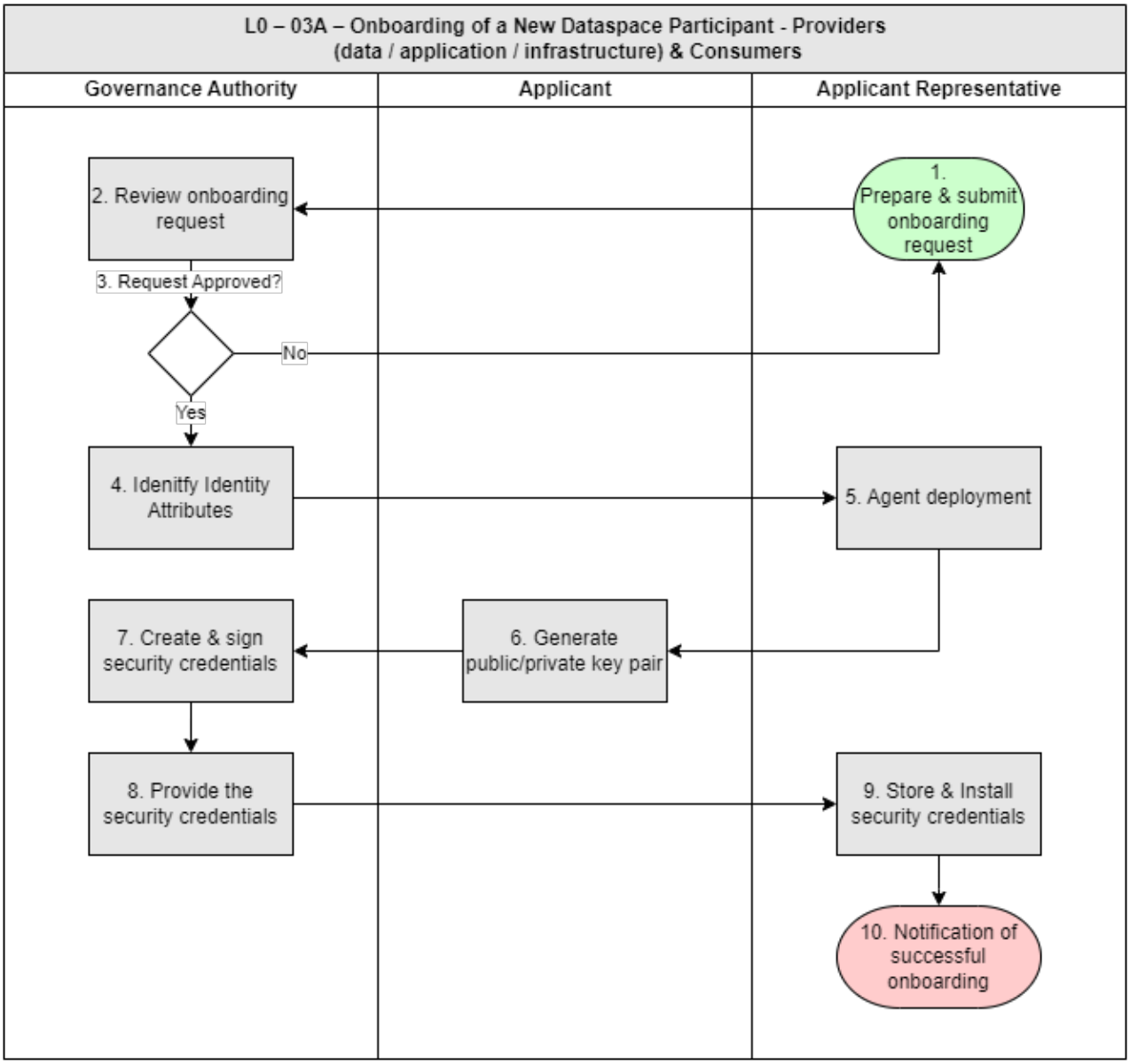
ASSUMPTIONS AND DEFINITIONS

DATASPACE APPLICANTS	Are assumed to be organisations, not individual person.
APPLICANT REPRESENTATIVE	Person/people acting on behalf of the applicant, is assumed to be a member of the Applicant organisation's directory.
SIMPL-AGENT	A local gateway enabling interaction with the dataspace.
DATASPACE SPECIFICATIONS	The document(s) describing the dataspace's objectives, candidature criteria and requirements applicable to an organisation for onboarding must be developed and made available to a potential applicant (e.g., website publication).

2.2 System Operations UML

The UML diagram, in this case, should be the same to the one provided by SIMPL-OPEN. The operations that MASA must perform correspond to those of the Applicant. MASA and other WP5 providers must complete the onboarding process on SIMPL-OPEN and provide the necessary information. This onboarding can be carried out through API or User Interface (UI) provided by SIMPL-OPEN.

2.2.1 UML Business Process Diagram



2.2.2 UML Diagram Commentary

1. **Prepare & submit onboarding request:** This initial step involves the Applicant Representative preparing a comprehensive application to participate to the dataspace, gathering the required information based on the documentation made available by the Governance Authority (GA). After the preparation of the onboarding request, the Applicant Representative fills in the forms and provides any other documents that may be mandatory (following the rules that are defined by the GA) to the Governance Authority for review.

2. **Review onboarding request:** After receiving the onboarding request, the GA starts the review process. It verifies the Applicant's onboarding request against a predefined set of criteria and the alignment with the dataspace objectives. The review process of the onboarding request can be either manually or automatically done by the GA.

3. **Request Approved?:** As an outcome of the review in step 2, the GA approves or rejects the request. If the request is approved, the process continues to the identification of Identity Attributes in step 4. In case deficiencies are found, the Applicant shall have the possibility to address them and start over the process from Step 1.

4. **Identify Identity Attributes:** After the approval the GA identifies the relevant Identity Attributes of the Applicant that will be used for authentication of the Applicant.

5. **Agent deployment:** If the application is approved, the Applicant Representative downloads the minimal set of modules from Simpl-Open that are required to be operative. The Applicant Representative then deploys and configures the modules on the Applicant's infrastructure to establish the needed environment for participating within the dataspace.

6. **Generate public/private keypair:** The Applicant's agent generates a public/private key pair to enable encrypted communications and data integrity within the dataspace. The private key is securely stored in the agent. The Applicant Representative shares the public key with the Governance Authority to request signed security credentials.

7. **Create & sign security credentials:** The Governance Authority creates and signs digital security credentials (e.g., x.509 certificates) that incorporate the Applicant's public key. These credentials serve as proof of identity and are validated through the issuance of certifications by the Governance Authority. The certifications ensure that the credentials are securely linked to the correct entity.

8. **Provide the security credentials:** The Governance Authority provides the signed security credentials to the Applicant. The security credentials are essential to ensure secure operations within the dataspace.

9. **Store & install security credentials:** The Applicant Representative stores and installs the signed identity security credentials in its Simpl-Open Agent.

10. **Notify the Applicant Representative of the Successful Onboarding:** The Applicant Representative is notified that they are now fully onboarded to the Dataspace and from now on are a Participant.

2.2.3 Data Exchange (WIP AND REVISION)

The providers (e.g. MASA) must submit the specifications requested by Governance Authority.

DATASPACE SPECIFICATIONS
• Dataspace's objectives • Identity attributes • Participant type (*DATA PROVIDER*)

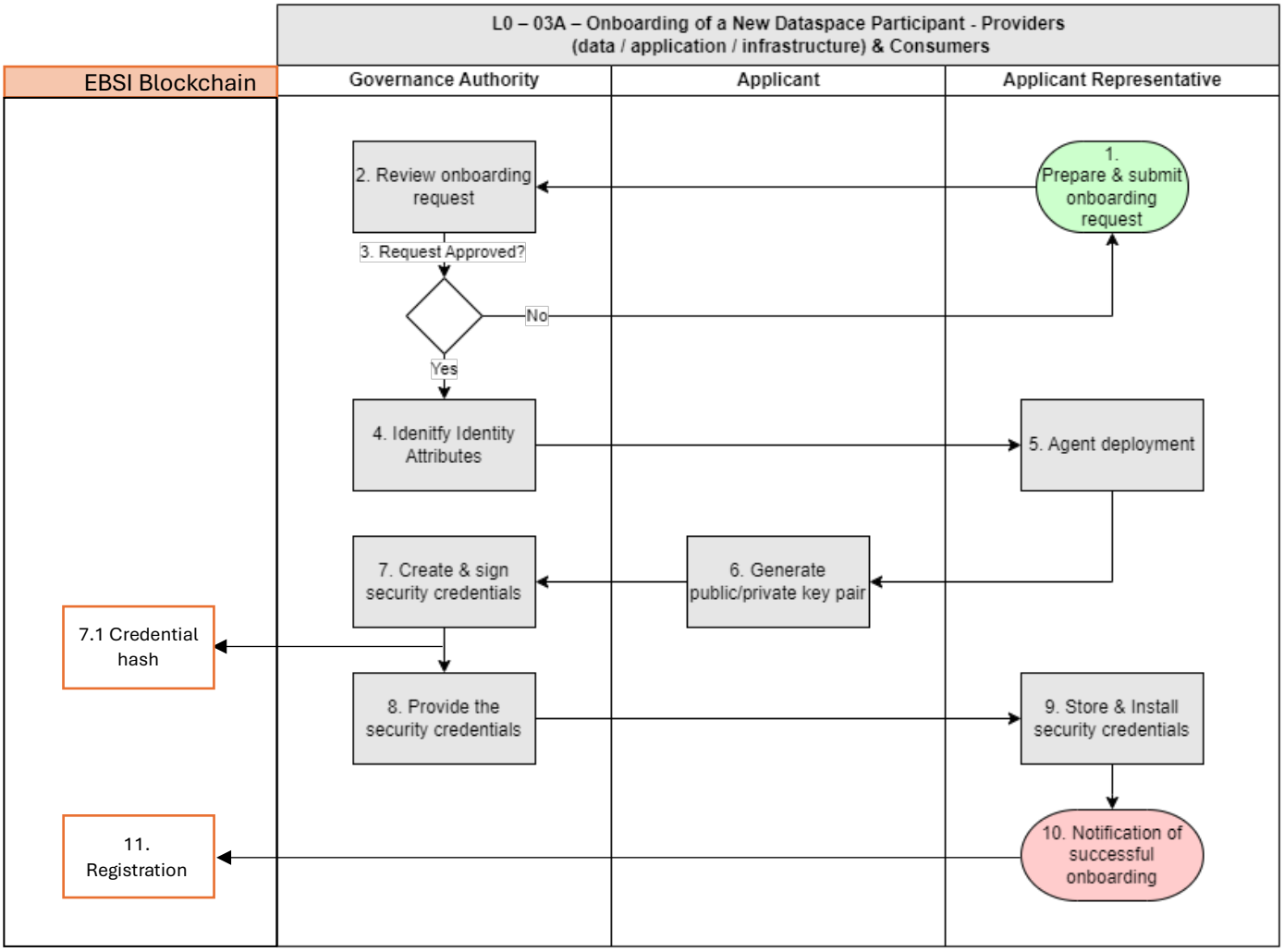
2.3 Smart Contract (WIP AND REVISION)

The process already includes robust security elements (key pair generation, security credentials, formal approval workflow), but as showed in the diagram in the next section, blockchain should enhance this in specific areas:

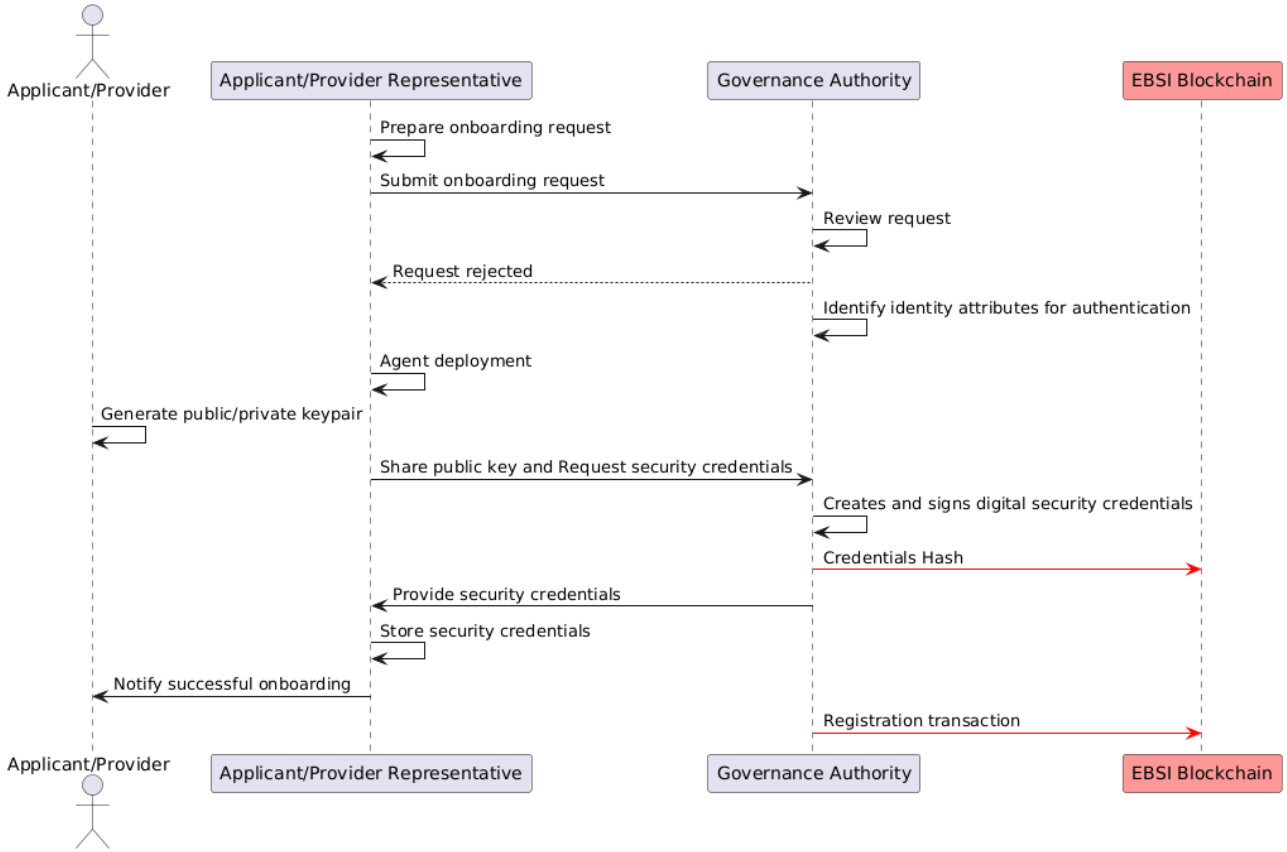
1. **After Step 7-8:** The signed security credentials represent a perfect blockchain use case. Recording a ***hash*** of these credentials on the blockchain would create an immutable verification layer.
2. **At Step 10:** The "Notification of successful onboarding" could trigger a blockchain transaction that officially registers the provider authorization.

In this case the blockchain could be serve as a complementary verification layer for the credentials and authorization status.

2.3.1 Smart contract flow (WIP & Temporary schema)



2.3.1.1 Sequence Diagram



Sequence Diagram – Onboarding process (created using Plant UML) *Click here to see/modify code*

2.3.2 Smart contract implementation (WIP AND REVISION)

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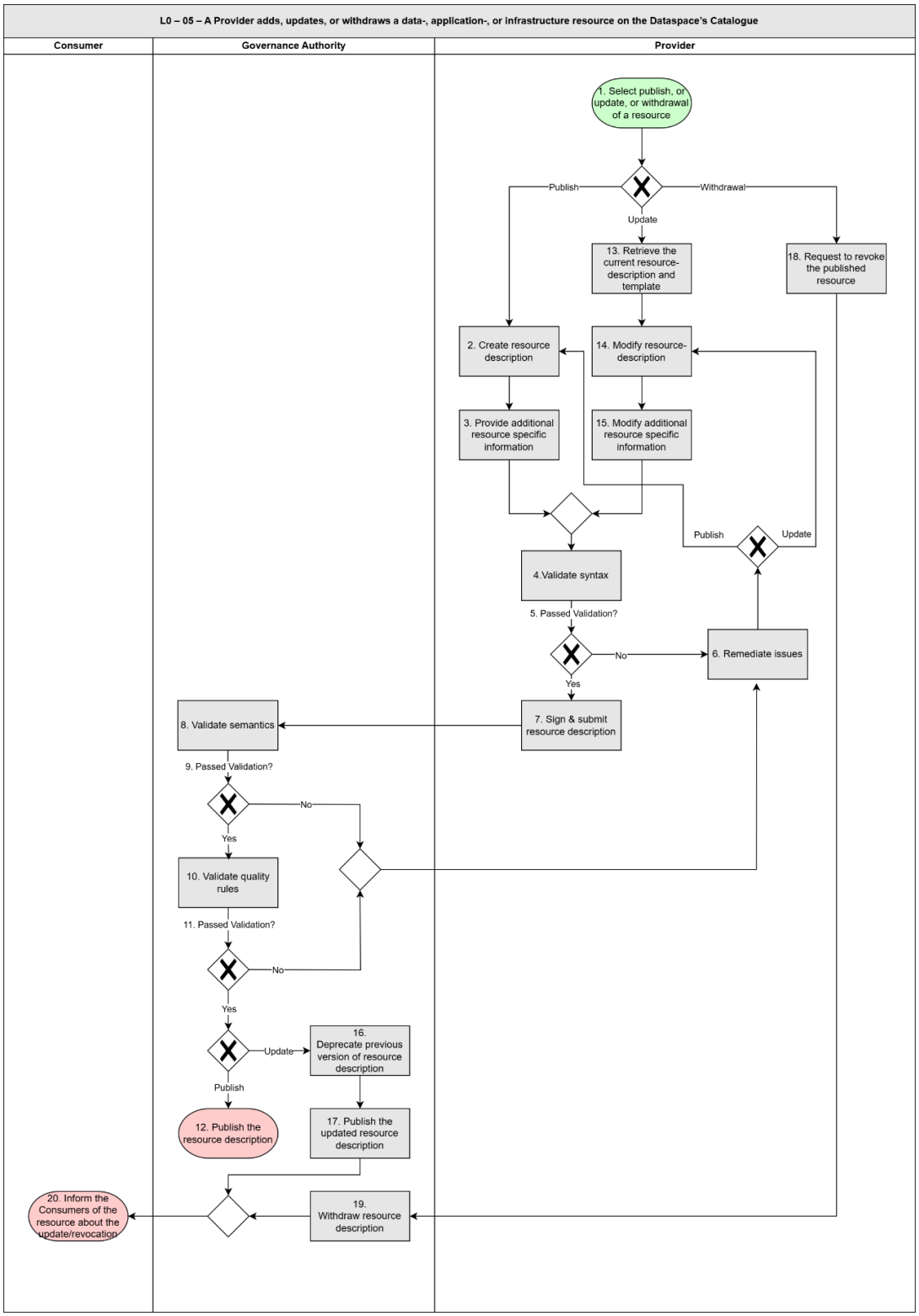
3. FUNCTION NAME: ADD_DATASET

3.1 Short Description

The documentation of the ADD process refers to the business process 5 of SIMPL_OPEN, available at the following link and shown in the following image. *LINK BP5*



In the following lines we ***focus only on the ADD operation*** to facilitate comprehension, make the various steps more understandable, and identify the various interactions with the blockchain depending on the type of operation. The ADD process describes the publication of resources on the dataspace's catalogue. Providers supply Resource Descriptions which may include access policies, usage policies and Usage Contract templates.
Resource Descriptions, in Use-Case#1, refer to data. These descriptions are validated for the correctness of syntaxes, semantics, and compliance with quality standards defined by the Governance Authority. Providers receive feedback on these validations. Once validated, the descriptions are published to the catalogue, making resources discoverable and accessible by other dataspace Participants.

Note 1: subsequently, the various operations of the business process will particularly refer to UC#1 and therefore to one of the main providers in this use case, namely MASA. MASA represents the operational actor and the data provider in use case #1.

Note *2:* it is important to consider that Resource Descriptions are immutable because they have unique signatures assigned at the time of creation. This immutability ensures the integrity and consistency of the resource, preventing unauthorised changes.

The following **actors** are involved in the current process:

- Governance Authority / *Simpl-Open*
- Provider / *MASA*

PREREQUISITES

The following prerequisites must be met to enable the process to occur:

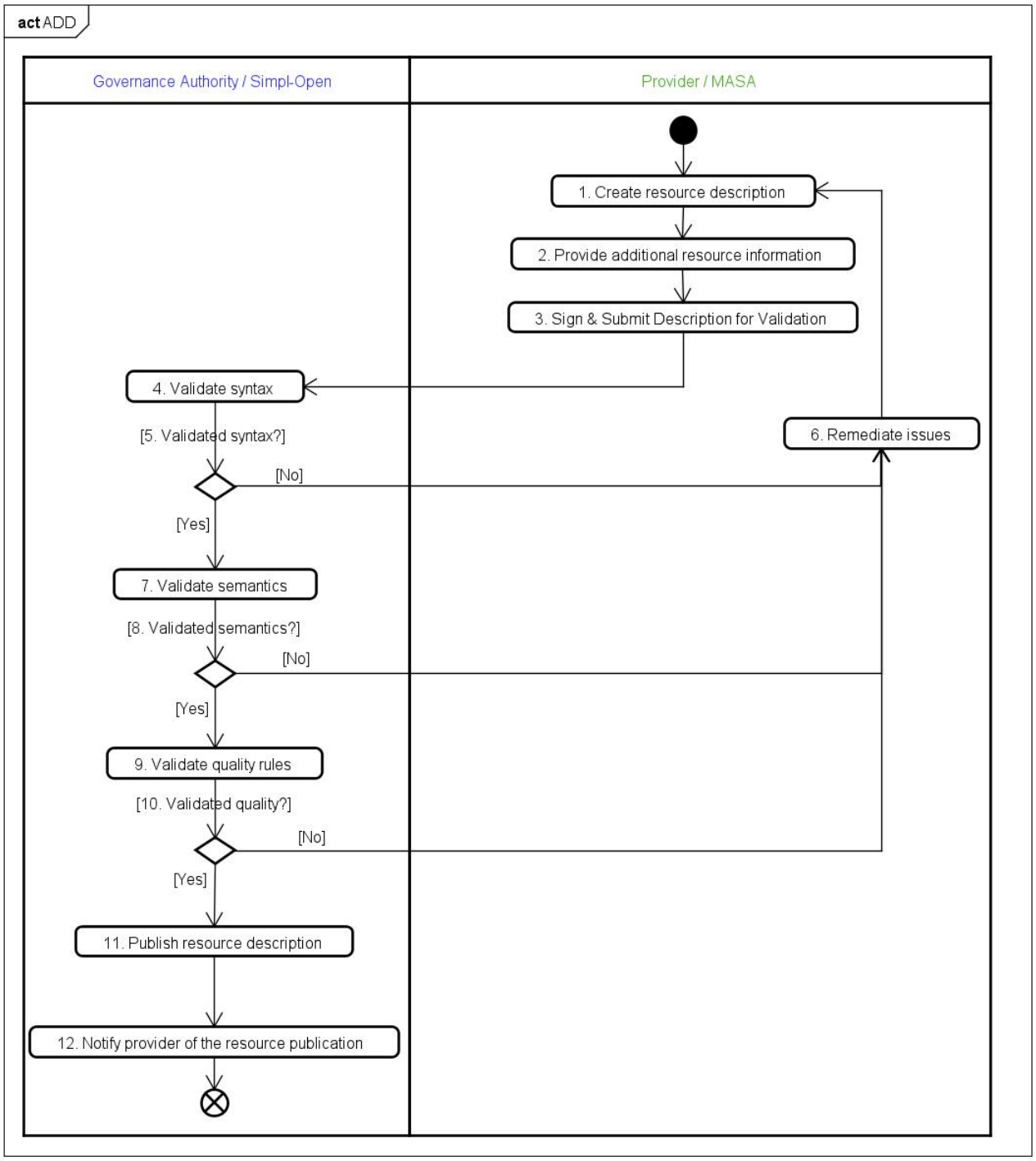
Provider Onboarded: Before the Provider can add any resources to a dataspace, they should have successfully completed the onboarding business process (*Business Process – 3A*).

Schemas Provided: The Governance Authority has defined and made available schemas for the different types of resources descriptions. These schemas may refer to vocabularies also available in the dataspace. Moreover, the Governance Authority has defined quality rules that will add further restrictions on acceptable Resource Descriptions so that their quality is guaranteed.

Resources registered at data space connector: The data space connector is the component to control and ensure data exchange via the dataspace protocol (DSP). Hence the resource must be registered as asset first to define the contract negotiation id needed for the resource descriptions to start contract negotiation.

3.2 System Operations UML

3.2.1 UML Business Process Diagram



Activity Diagram – ADD process - Provider - Simpl-open (created using Astah software)

3.2.2 UML Diagram Commentary

1. **Create Resource Description:** The Provider (e.g. MASA) fetches schemas which describe the mandatory and recommended fields provided by the data Governance Authority. Based on the schema retrieved, the Provider can create the description of the resource.

2. **Provide additional resource specific information**: To fully describe data resources, additional resource-specific information is required. The specific details within the resource schema may vary depending on the resource type. For example: Data Resources may require: Data Provenance, Format under which the data is distributed, Schema of the dataset, Related datasets, Target users, Data Quality, Encryption, and Anonymisation /Pseudonymisation. Additionally, Providers may choose to assign further information, such as access policies, usage policies, Usage Contract template details, and more, to these resources.

3. **Sign & Submit Resource Description for Validation**: The description is digitally signed by the Provider and sent to the Governance Authority for validation.

4. **Validate syntax:** The Resource Description is validated on the correctness of the included syntaxes (e.g., specific format and structure rules).

5. **Validated syntax?:** If the syntax is successfully validated, then the process continues to step 7. If syntax does not pass validation, then the process returns to step 6.

6. **Remediate issues:** After the Provider receives the error report, the Provider can correct the errors, and reverts to step 1 for the 'Create Resource Description' process.

7. **Validate semantics**: The Resource Description is semantically validated against the relevant vocabularies and constraints defined in the schema by the Governance Authority.
8. **Validated semantics?:** If semantics is successfully validated, then the process continues to step 9. If semantics does not pass validation, then the process returns to step 6.
9. **Validate quality rules**: The Resource Description is validated against the quality rules (and compliance-specific rules for the applications) defined by the Governance Authority.
10. **Validated quality rules?:** If the quality is successfully validated, then the process continues to step 11 in case a resource is newly published. If the quality does not pass validation, then the process goes to step 6.
11. **Publish resource description**: Once all validations are successfully completed, the Resource Description of the resource is published on the Dataspace Catalogue along with its associated metadata, making it discoverable to other Participants (*Business Process – 06*).

12. **Notify provider of the resource publication.**

3.2.3 Data Exchange (WIP AND REVISION)

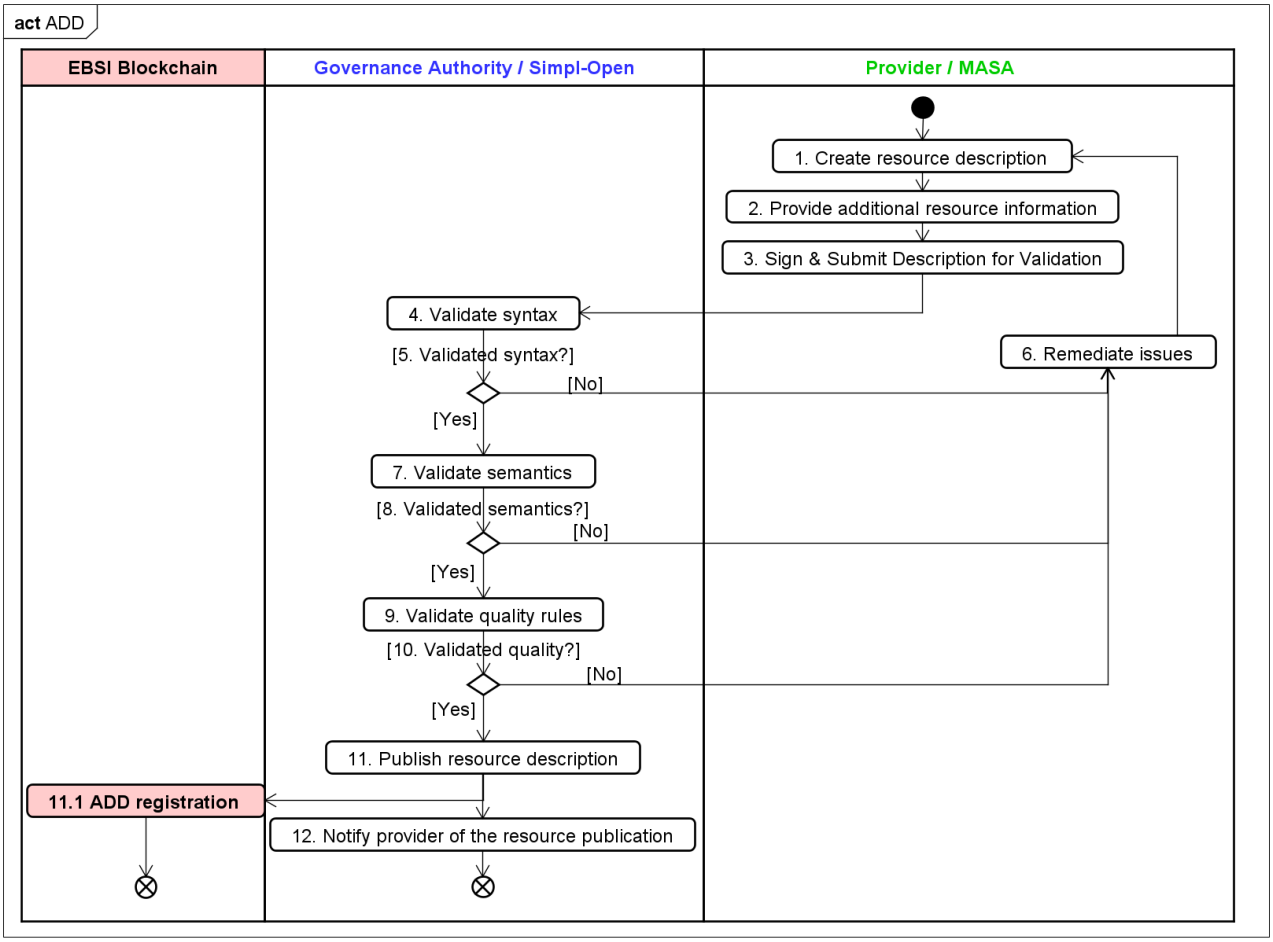
In this section, an example of which data can be exchanged in the ADD operation is showed.

DATASPACE SPECIFICATIONS
• Resource_hash • Data_provenance • Operation_type • Authentication_type • Timestamp • Resource_position • Operation_metadata • User_metadata • Signature • Resource_format • Resource_size • Schema_of_the_dataset • Related_datasets • Target users • Encryption • Anonymisation/Pseudonymisation

3.3 Smart Contract (WIP AND REVISION)

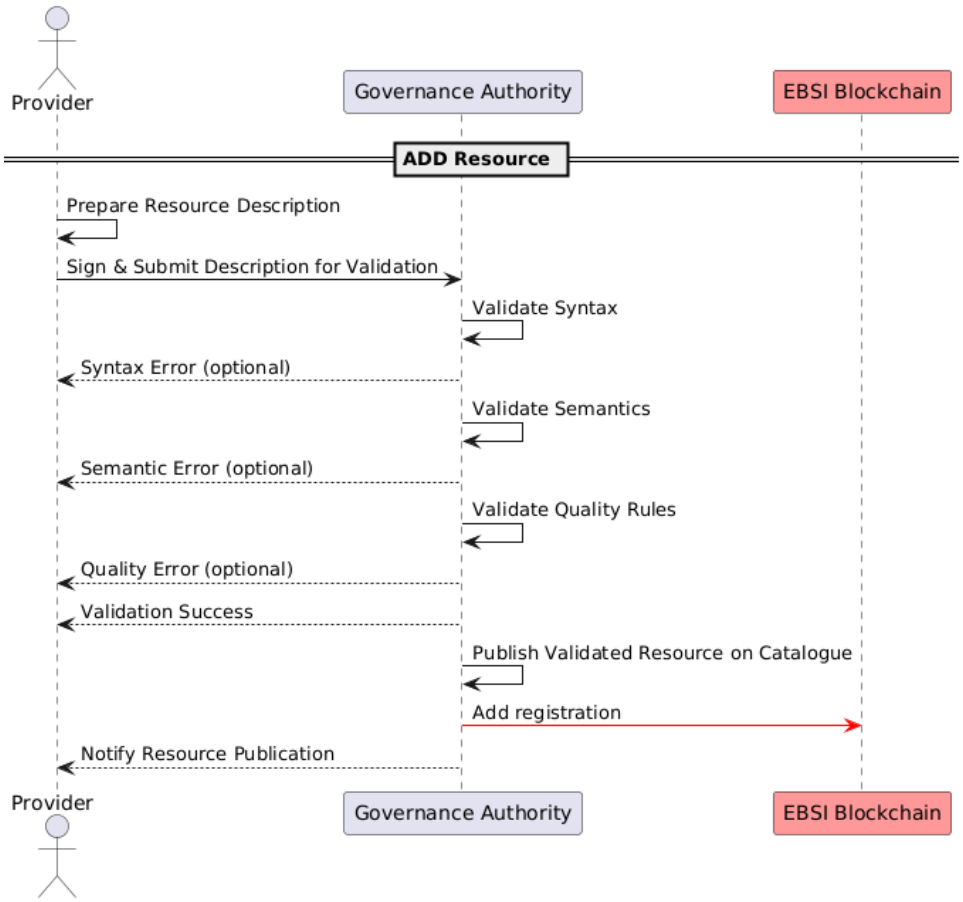
In the diagram shown in the next section, **Step 11.1 (ADD registration):** after the resource is published on the catalogue, a blockchain transaction can officially register the successful ADD operation.

3.3.1 Smart contract flow (WIP AND REVISION)



Activity Diagram – ADD process - Provider - Simpl-open - Blockchain (created using Astah software)

3.3.1.1 Sequence Diagram and detailed data specifications



Sequence Diagram – ADD process (created using Plant UML). Click here to see/modify code

In the case of UseCase#1, where the provider is MASA, the resources to be added to the catalogue consist in **data**.
An initial set of fields to be included in the data exchanged during the *ADD resource* process is outlined in the section **3.2.3 Data Exchange (WIP)**.

Specifically considering MASA provider, the *Operation_metadata* will be in JSON format and could include the following fields:

- Application_id
- Timestamp
- Event_data

The last field, *event_data*, will be event-specific, meaning it depends on the type of event to be added to the dataspace. For instance, MASA may share data related to

- Collisions
- Alerts sent to road users
- Traffic violations
- Road-users GPS

3.3.2 Smart contract implementation (WIP)

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4. FUNCTION NAME: UPDATE/ADD_NEW_VERSION_DATASET

4.1 Short Description

The documentation of the UPDATE/ADD_NEW_VERSION process refers to the dedicated part included in the business process 5 of SIMPL_OPEN, available at the following link. *LINK BP5*

When updating a resource, it is important to consider that Resource Descriptions are immutable because they have unique signatures assigned at the time of creation. This immutability ensures the integrity and consistency of the resource, preventing unauthorised changes.

As a result, in Simpl-open dataspace, updating a resource involves deprecating the old version and uploading a new version with a new immutable signature.

Instead, in the blockchain the deprecation of a resource is impossible due to the immutability feature of blockchain itself, thus the update operation consists in the registration of an ADD transaction for the new version of the resource. The new block associated with the update transaction is linked to the previous one (where the hash of the old version of the resource was stored).

In line with these considerations, the name of the current function is renamed in UPDATE/ADD_NEW_VERSION.

The following **actors** are involved in the current process:

- Governance Authority / *Simpl-Open*
- Provider / *MASA*

PREREQUISITES

The following prerequisites must be met to enable the process to occur:

Provider Onboarded: Before the Provider can add any resources to a dataspace, they should have successfully completed the onboarding business process (*Business Process – 3A*).

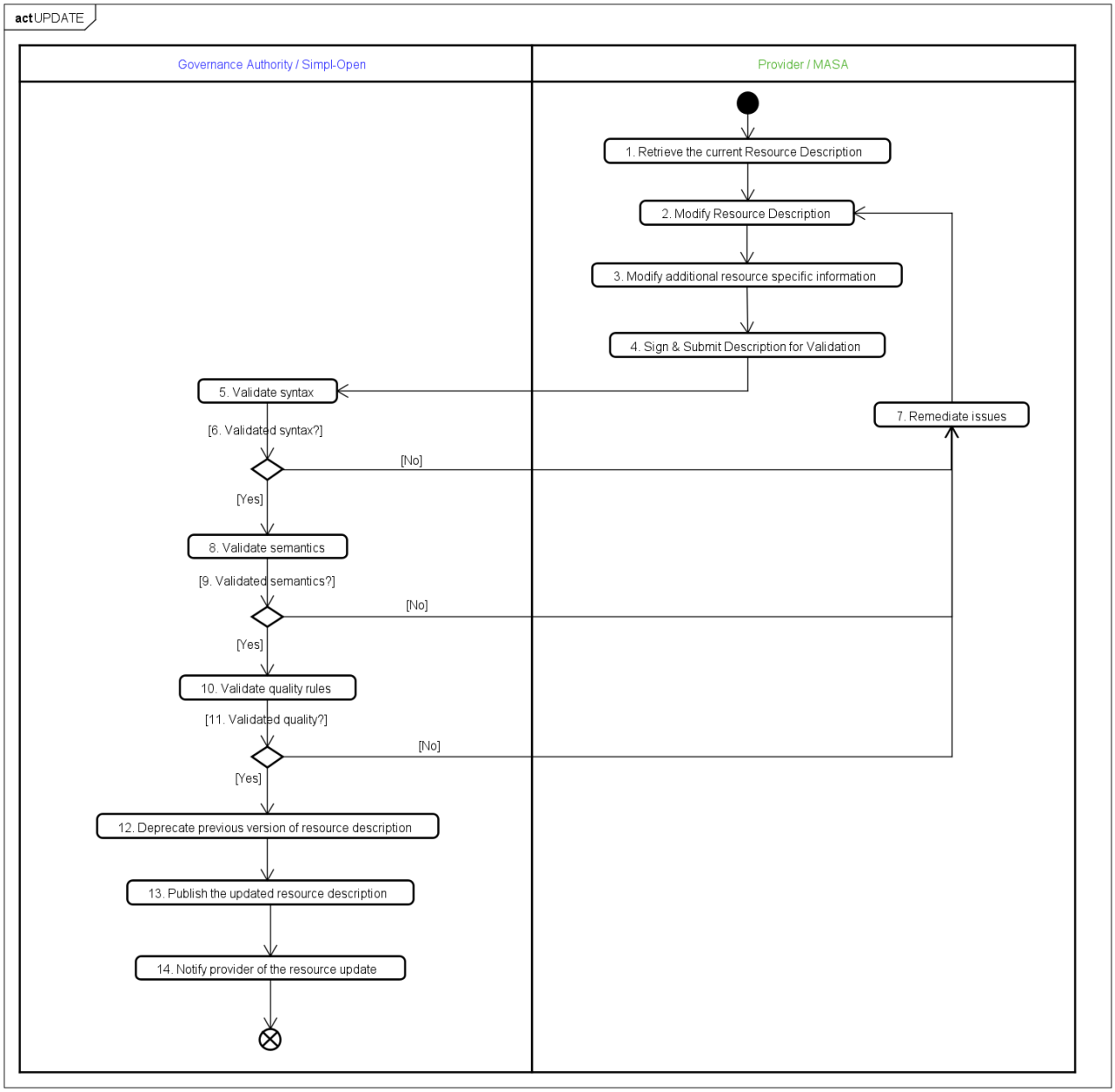
Schemas Provided: The Governance Authority has defined and made available schemas for the different types of resources descriptions. These schemas may refer to vocabularies also available in the dataspace. Moreover, the Governance Authority has defined quality rules that will add further restrictions on acceptable Resource Descriptions so that their quality is guaranteed.

Resources registered at data space connector: The data space connector is the component to control and ensure data exchange via the dataspace protocol (DSP). Hence the resource must be registered as asset first to define the contract negotiation id needed for the resource descriptions to start contract negotiation.

Previous version of the resource already stored in the dataspace.

4.2 System Operations UML

4.2.1 UML Business Process Diagram



Activity Diagram – UPDATE/ADD_NEW_VERSION process - Provider - Simpl-open (created using Astah software)

4.2.2 UML Diagram Commentary

1. **Retrieve the current Resource Description**: The Provider (e.g. MASA) requests the current Resource Description and the current Resource Description Schema from the registry.
2. **Modify Resource Description:** Based on the schema retrieved, the Provider can modify the Resource Description. The Provider will always use the latest version of the schema even if it is different from the schema of the original Resource Description.
3. **Modify additional resource specific information:** The Provider can modify various types of information that is provided in the description, which are described in *additional resource specific information* here.
4. **Sign & submit modified Resource Description:** The modified description is digitally signed by the Provider and sent to the Governance Authority for validation.
5. **Validate syntax:** The modified Resource Description is validated on the correctness of the included syntaxes (e.g., specific format and structure rules).
6. **Validated syntax?:** If the syntax is successfully validated, then the process continues to step 8. If syntax does not pass validation, then the process returns to step 7.
7. **Remediate issues:** After the Provider receives the error report, the Provider can correct the errors, and reverts to step 2 for the 'Modify Resource Description' step.
8. **Validate semantics**: The Resource Description is semantically validated against the relevant vocabularies and constraints defined in the schema by the Governance Authority.
9. **Validated semantics?:** If semantics is successfully validated, then the process continues to step 10. If semantics does not pass validation, then the process returns to step 7.
10. **Validate quality rules**: The Resource Description is validated against the quality rules (and compliance-specific rules for the applications) defined by the Governance Authority.
11. **Validated quality rules?:** If the quality is successfully validated, then the process continues to step 12. If the quality does not pass validation, then the process goes to step 7.
12. **Deprecate previous version of resource description:** The Governance Authority automatically deprecates the current Resource Description after the new Resource Description is fully validated.
13. **Publish the updated resource description:** The updated Resource Description of the resource is published on the dataspace catalogue, making it discoverable to other Participants (*Business Process – 06*).
14. **Notify provider of the resource update**.

4.2.3 Data Exchange (WIP AND REVISION)

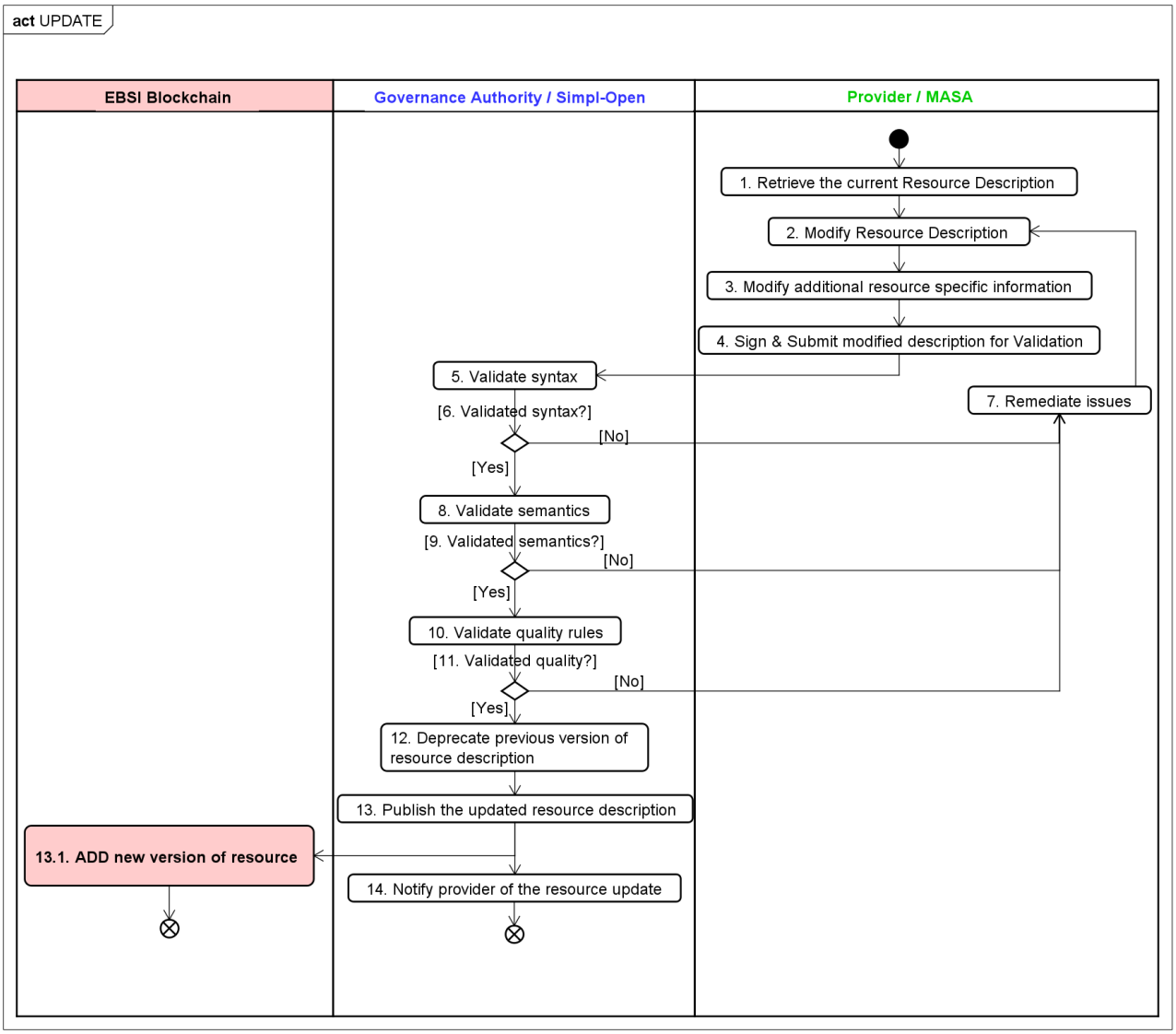
DATASPACE SPECIFICATIONS
• Resource_hash_updated_version • Resource_hash_previous_version • Data_provenance • Operation_type • Authentication_type • Timestamp • Resource_position • Operation_metadata • User_metadata • Signature • Resource_format • Resource_size • Schema_of_the_dataset • Related_datasets • Target users • Encryption • Anonymisation/Pseudonymisation

Resource_hash_updated_version: this field stores the hash of the new version of the resource, ensuring version control and integrity of data.
Resource_hash_previous_version: this field represents the hash of the previous version of the resource, allowing the system to maintain a chain of updates.

4.3 Smart Contract (WIP AND REVISION)

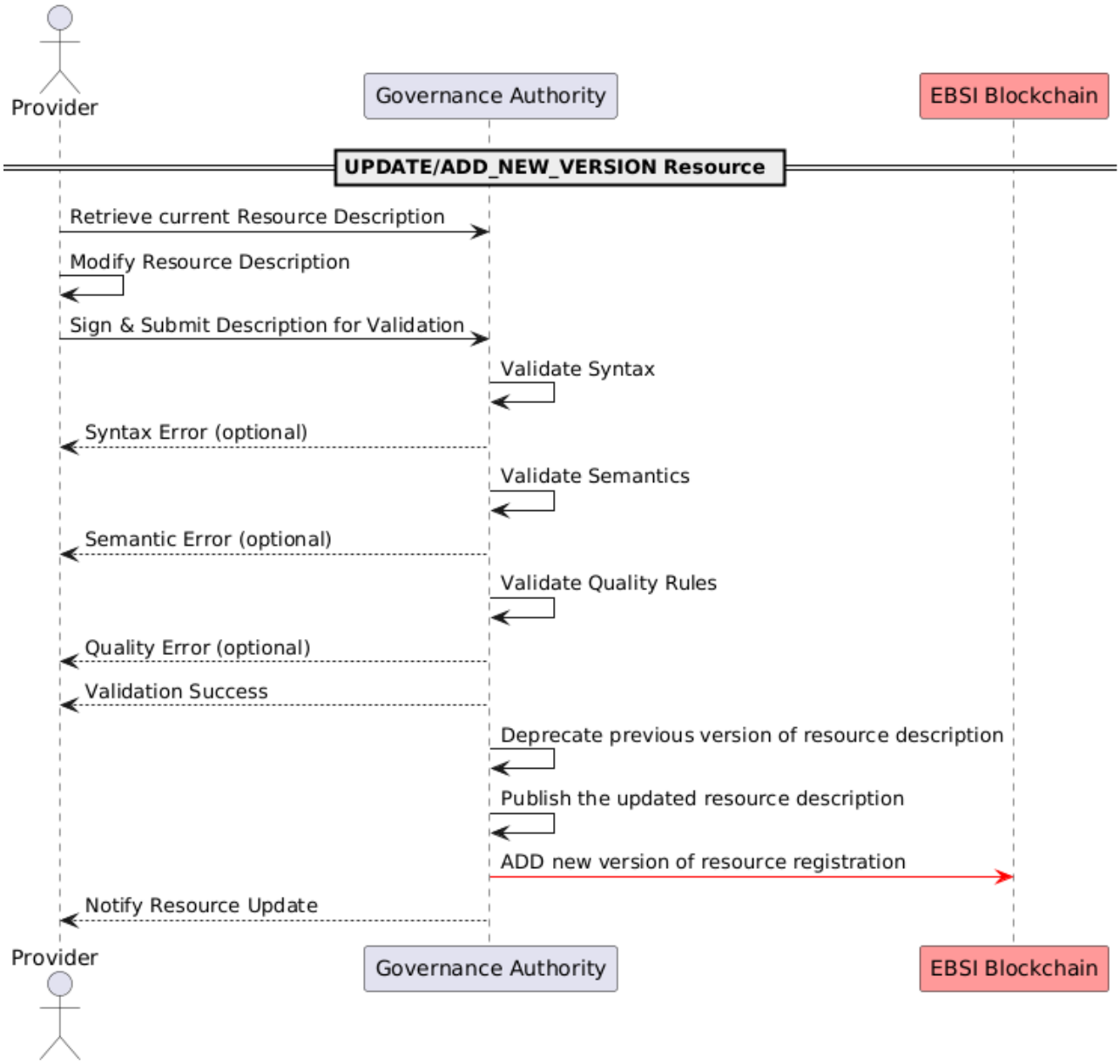
In the diagram shown in the next section, step **13.1 (ADD new version of the resource)**: after the new version of the resource is stored in the dataspace, the update operation is registered in the blockchain as an ADD transaction for the new version of the resource. The new block associated with the update transaction is linked to the previous one through the hash of the old version of the resource. This approach ensures the traceability of all resource's versions.

4.3.1 Smart contract flow (WIP)



Activity Diagram – UPDATE/ADD_NEW_VERSION process - Provider - Simpl-open - Blockchain (created using Astah software)

4.3.1.1 Sequence Diagram and detailed data specifications



Sequence Diagram – UPDATE/ADD_NEW_VERSION process (created using Plant UML). Click here to see/modify code

4.3.2 Smart contract implementation (WIP)

....

5. FUNCTION NAME: WITHDRAWAL/REVOKE_DATASET

5.1 Short Description

The documentation of the WITHDRAWAL/REVOKE process refers to the dedicated part included in the business process 5 of SIMPL_OPEN, available at the following link. *LINK BP5*

In Simpl-open dataspace, withdrawing a resource involves the removal operation from the dataspace.

Instead, in the blockchain the resource withdrawal is impossible due to the immutability feature of blockchain itself, thus the withdrawal operation consists in the registration of a REVOKE transaction to completely hinder the access to the resource. The block associated with the revoke transaction is linked to the previous one (where the resource hash was stored).

In line with these considerations, the name of the current function is renamed in WITHDRAWAL/REVOKE.

The following **actors** are involved in the current process:

- Governance Authority / *Simpl-Open*
- Provider / *MASA*

PREREQUISITES

The following prerequisites must be met to enable the process to occur:

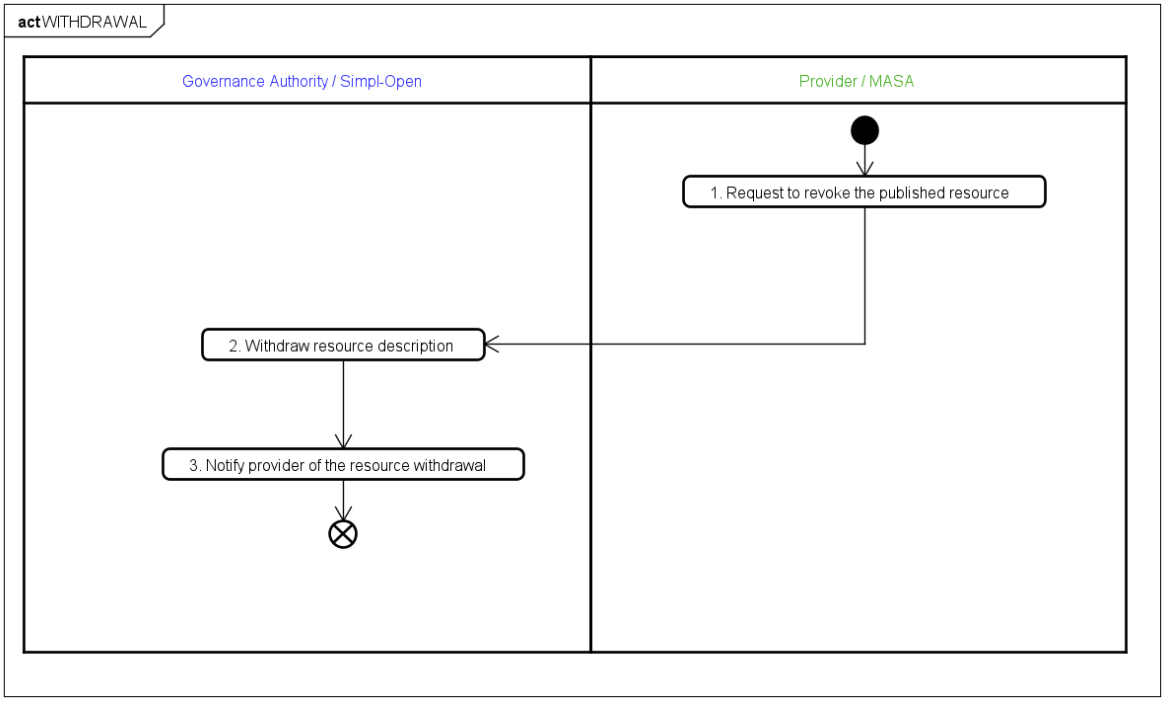
Provider Onboarded: Before the Provider can modify any resources to a dataspace, they should have successfully completed the onboarding business process (*Business Process – 3A*).

Resources registered at data space connector: The data space connector is the component to control and ensure data exchange via the dataspace protocol (DSP). Hence the resource must be registered as asset first to define the contract negotiation id needed for the resource descriptions to start contract negotiation.

Resource already stored in the dataspace.

5.2 System Operations UML

5.2.1 UML Business Process Diagram



Activity Diagram – WITHDRAWAL/REVOKE process - Provider - Simpl-open (created using Astah software)

5.2.2 UML Diagram Commentary

1. **Request to revoke the published resource:** The Provider signals the Governance Authority to revoke the published resource from the catalogue.
2. **Withdraw resource description:** The Governance Authority withdraws the resource description from the Dataspace's Catalogue.
3. **Notify provider of the resource withdrawal.**

5.2.3 Data Exchange (WIP)

DATASPACE SPECIFICATIONS
• Resource_hash_revoked • Resource_hash_previous_version • Data_provenance • Operation_type • Authentication_type • Timestamp

- Resource_position
- Operation_metadata
- User_metadata
- Signature
- Resource_format
- Resource_size
- Schema_of_the_dataset
- Related_datasets
- Target users
- Encryption
- Anonymisation/Pseudonymisation

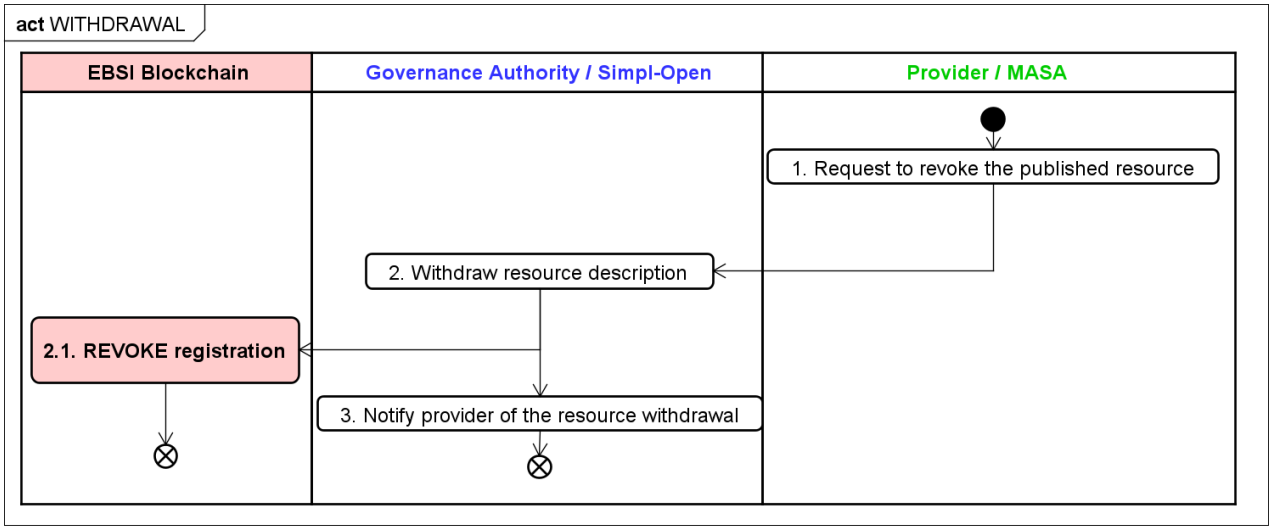
Resource_hash_revoked: it stores the hash of the resource that has been withdrawn from the dataspace, ensuring traceability of all the resource's states up to the notorization of the revoked access to the resource on the blockchain.

Target users: this field refers to authorized accesses to the transaction block, so in the WITHDRAWAL/REVOKE operation, all the access are hindered, except for the provider and governance authority accesses.

5.3 Smart Contract

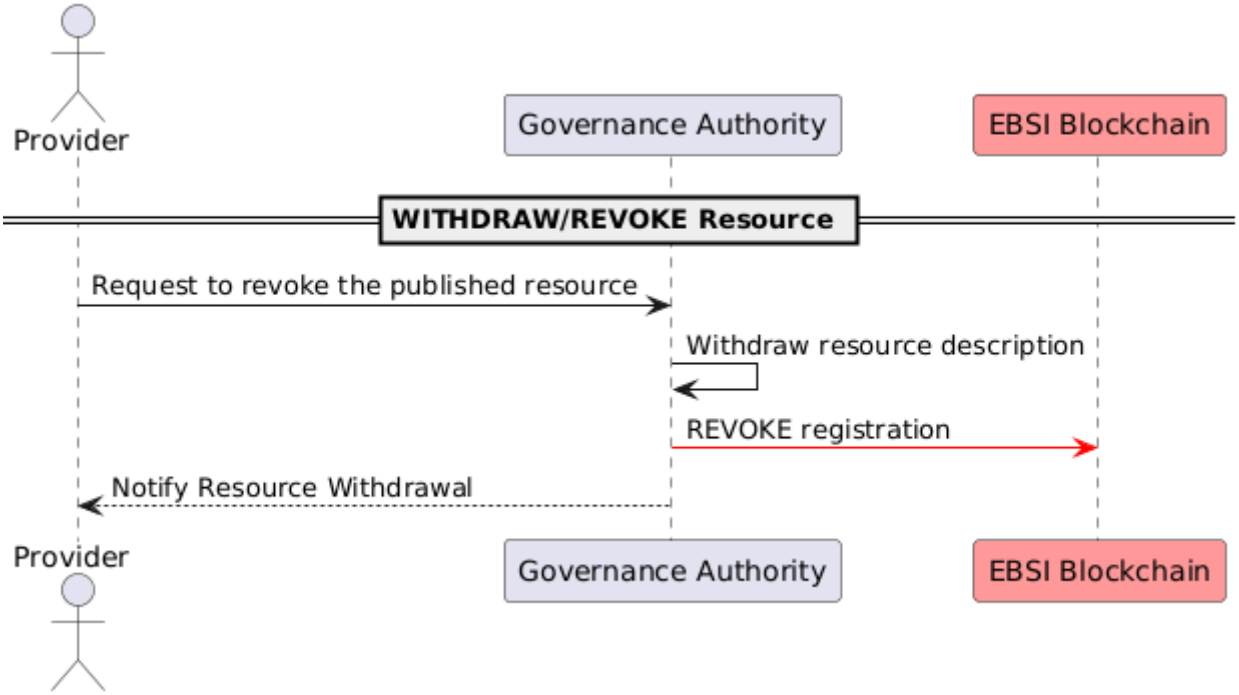
In the diagram shown in the next section, step **2.1 (REVOKE registration)**: when the provider withdraws a resource in the dataspace, in the blockchain this operation consists in the creation of a new block which is linked to the previous one associated with the last operation on the resource (e.g. ADD or UPDATE). This new block notarizes the revocation of access to the resource. As a result, all access to the resource is revoked, except for the provider and governance authority accesses. Thus, this ensures that the traceability and data integrity are maintained, and the history of the resource is preserved.

5.3.1 Smart contract flow (WIP)



Activity Diagram – WITHDRAWAL/REVOKE process - Provider - Simpl-open - Blockchain (created using Astah software)

5.3.1.1 Sequence Diagram and detailed data specifications



Sequence Diagram – WITHDRAWAL/REVOKE process (created using Plant UML). Click here to see/modify code

5.3.2 Smart contract implementation (WIP)

....

6. PROOF OF CONCEPT (POC): BLOCKCHAIN_IMPLEMENTATION

6.1. Introduction

This chapter describes the implementation of a Proof of Concept (POC) developed to simulate and test the core functionalities of a blockchain-integrated data exchange architecture. The aim is to validate key technical assumptions and evaluate operational feasibility before integration into a real use case.

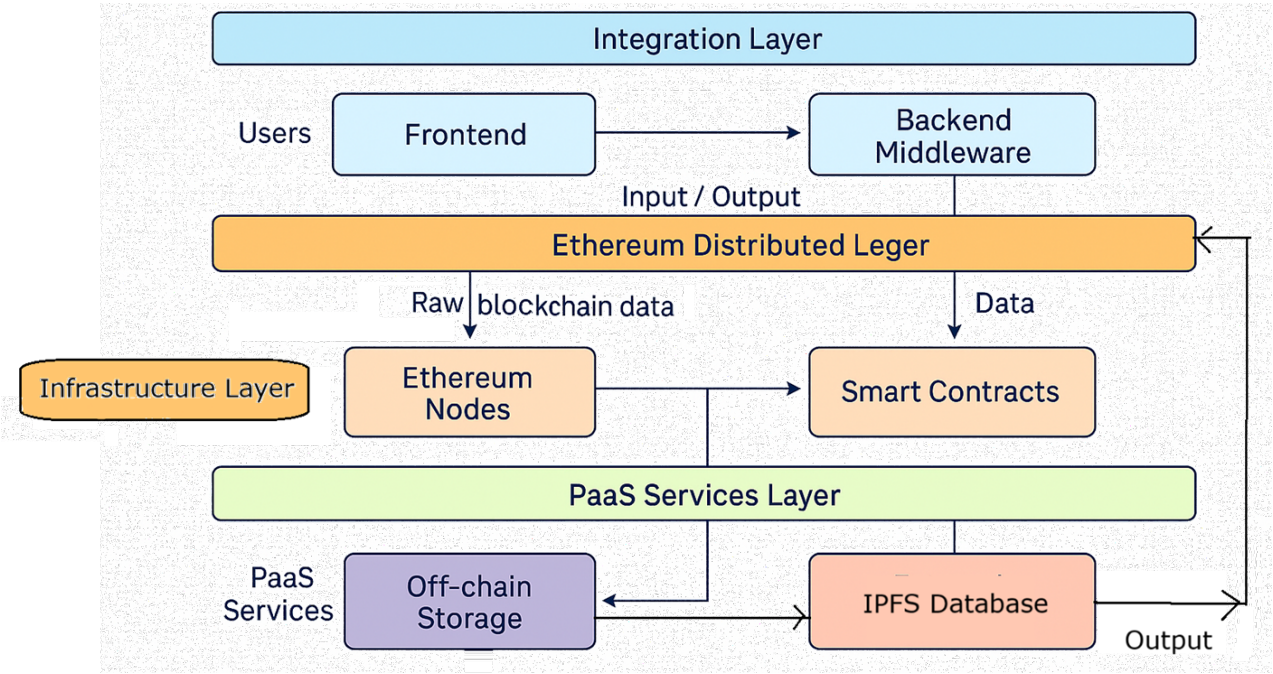
6.2. General Architecture Overview

In the proposed architecture, the EBSI Blockchain is integrated into the Simpl-Open framework, following a layered design that ensures modularity, scalability, and clean separation of concerns.

At the Integration Layer, users interact with the system through a frontend interface, while backend middleware manages secure and reliable communication with the EBSI blockchain. These inputs are passed to the Distributed Ledger, which acts as a decentralized database — storing both on-chain transaction data and references to off-chain resources such as externally hosted files or metadata.

At the Infrastructure Layer, EBSI nodes handle consensus, transaction validation, and smart contract execution. Smart contracts produce outputs like confirmed transactions and event logs. For large or sensitive data, the system uses off-chain storage, with content linked via cryptographic hashes to ensure integrity.

Finally, at the PaaS Services Layer, external storage solutions manage long-term file retention and off-chain metadata, while outputs — including user-relevant data and smart contract results — flow back to the application layer for display and interaction.



General architecture overview: components and layers

6.3. Objective of the POC

The primary goal of the POC is to simulate blockchain operations related to the various business process described on the previous chapters (BP03, BP05_A etc.).

This first step of implementation includes:

- Simulating smart contracts
- Testing Ethereum-based transactions
- Testing the block creation for each operation

6.4. Tools Used

The POC implementation leverages the following tools:

- **Ganache**: A local Ethereum blockchain simulator used for quick and controlled testing.
- **Truffle**: A development framework for compiling, deploying, and testing smart contracts.

While Ganache was chosen for its ease of use and local testing capabilities, we remain open to future migration to more robust environments such as Hyperledger Besu to align with project-wide decisions.

6.5. Implementation Details

The POC focused on the ADD operation as the initial business process. The implementation followed these steps:

1. **Development of a Simplified Smart Contract**
   a. The smart contract was written in Solidity.
   b. It manages the registration of new resources and logs the operation on the blockchain.
2. **Components of the Contract**
   a. event ResourceAdded(...): Emits every time a resource is added. Captures hashed ID, provider address, and metadata hash.
   b. mapping (bytes32 => bool) exists: Prevents duplicates by keeping track of registered resources.
   c. function addResource(...): Validates uniqueness, updates the mapping, and emits the event.
3. **Deployment and Testing**
   a. The contract was compiled and deployed locally using Truffle + Ganache.
   b. Test transactions were executed to simulate actual usage (see slides 11–12).
   c. Each transaction led to the creation of a new block containing the resource data.

resource_add_registry.test.js | ResourceAddRegistry.sol

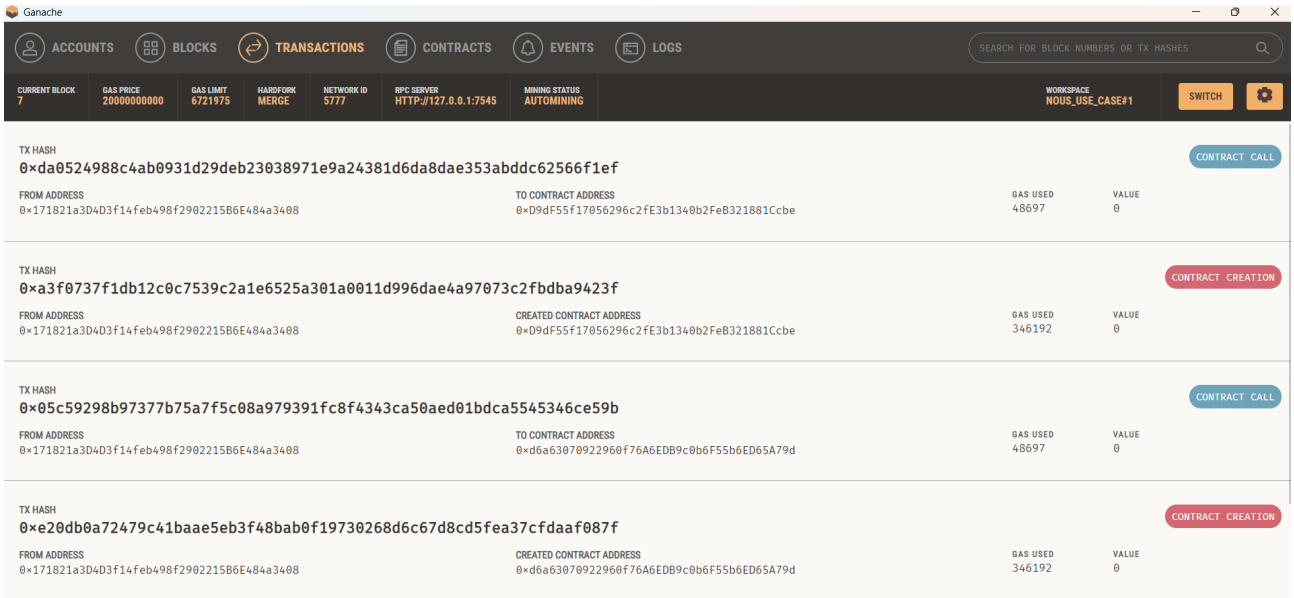
contracts > ResourceAddRegistry.sol > ...

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract ResourceAddRegistry {
    event ResourceAdded(bytes32 indexed resourceId, address provider, string metadataHash);

    mapping(bytes32 => bool) public exists;

    function addResource(bytes32 resourceId, string memory metadataHash) public {
        require(!exists[resourceId], "Resource already registered");

        exists[resourceId] = true;

        emit ResourceAdded(resourceId, msg.sender, metadataHash);
    }
}

```

Smart contract: Add_operation – Written in Solidity programme language

6.6. Results and Observations

- Ganache proved effective for initial validation.
- Transactions and blocks were correctly generated.
- The smart contract correctly prevented duplicates and logged every ADD operation.
- Future implementations will include MODIFY and REVOKE operations.



Ganache test: Transaction for Add_operation

6.7. Conclusion

The POC confirmed the technical feasibility of blockchain-backed business processes. Local testing with Ganache provided a reliable environment to validate key behaviours. However, full validation requires real integration with SIMPL-OPEN and access to its operational endpoints.

Testing SIMPL-OPEN remains a strategic priority to ensure that provider onboarding, authentication flows, and blockchain interactions work cohesively within the designed architecture.

A. APPENDIX N.1: GLOSSARY

TERMINOLOGY	DEFINITION
Add operation	Business Process that registers a new resource (dataset) within the system, typically including blockchain notarization.
Applicant	Dataspace Applicants are assumed to be organizations and not individual persons that are interested and proceed to the onboarding process on Simpl-Open.
Applicant Representative	Person/people acting on behalf of the applicant, is assumed to be a member of the Applicant organisation's directory.
BESU Hyperledger	An open-source Ethereum client designed for enterprise use, supporting both public and permissioned networks.
Blockchain	A distributed, immutable ledger that maintains an ever-growing chronological sequence of blocks containing transaction data.
Consumer	In the SIMPL-OPEN context, a consumer refers to any entity—whether an organization or an individual—that intends to access or use a published resource. Consumers interact with the dataspace to discover, request, and retrieve available resources.
DID – Decentralized IDentifier	A URL-based identifier consisting of a string of letters and numbers that identifies a subject, such as a natural person or legal entity, without revealing any meaningful information about them. These identifiers are most often used with Verifiable Credentials and associated with subjects in such a way that the verifiable credential can be easily ported from one repository to another without the need to reissue the credentials.
Data exchange	The process of transferring data between systems or organizations, often standardized and secured.
Dataspace: SIMPL-OPEN	A federated environment enabling secure and standardized data sharing between stakeholders, where SIMPL-OPEN acts as the orchestrating platform.

Distributed Ledger	A data structure replicated over multiple entities in a peer-to-peer network, whereby a consensus mechanism guarantees the consistency of the different copies of this data structure, without the need for central authority.
EBSI	European Blockchain Services Infrastructure; a blockchain network promoted by the EU for cross-border digital services and verifiable credentials.
Ethereum	A decentralized blockchain platform that enables smart contract execution and D-App development.
Ethereum Testnet	A testing environment for Ethereum developers to deploy and test smart contracts without using real Ether.
Ganache	A local Ethereum testnet blockchain used for testing and development of smart contracts.
Governance authority (GA)	The "Governance Authority" within the Simpl Programme framework refers to an entity that manages and controls access to data within a data space. This authority is responsible for tasks like validating participant onboarding, issuing and managing security credentials, and enforcing data access policies
Hash	A fixed-length output generated from an input of any size using a cryptographic function, used for data integrity and identification.
MASA	The main provider in Use Case #1 is responsible for submitting resource data into the SIMPL-OPEN architecture.
Metadata	Descriptive information about a resource or dataset, often used to aid in identification, categorization, or usage (e.g. name, title, file type, category, language, issuer, timestamp, etc).
Middleware	Software that connects different systems or applications, enabling communication and data exchange between them.

Onboarding operation	The process of registering and verifying a new data provider or user in the SIMPL-OPEN environment.
Oracle	A system that provides external data to smart contracts, enabling interaction with off-chain information.
PaaS	Platform as a Service; a cloud computing model that provides a platform for developers to build, deploy, and manage applications.
Resource	In Simpl-Open context, the word "resource" is a general term that includes data, application and infrastructure. Each provider should add a resource, and this resource should be a dataset, an application, or an entire infrastructure. Specifically, in UC#1, the term "resource" refers exclusively to data.
Revoke/withdraw operation	A business operation that removes access to a previously registered resource or revokes permissions. In Simpl-open dataspace, withdrawing a resource involves the removal operation from the dataspace. Instead, in the blockchain the resource withdrawal is impossible due to the immutability feature of blockchain itself, thus the withdrawal operation consists in the registration of a REVOKE transaction to completely hinder the access to the resource. The block associated with the revoke transaction is linked to the previous one (where the resource hash was stored).
Smart contract	A computer program published on a distributed ledger that automatically executes agreements when predetermined conditions are met. Due to the consensus mechanism of the ledger, no one should be able to unilaterally affect the correct execution of the smart contract code.
Solidity	A programming language for writing smart contracts on Ethereum and compatible blockchains.
UML activity diagram	A UML diagram that models the workflow or activities involved in a process.
UML sequence diagram	A UML diagram that shows how actors interact in a specific sequence to complete a process.

Update/New Add operation	A business process used to modify the information of an existing resource in the system. In Simpl-open dataspace, updating a resource involves deprecating the old version and uploading a new version with a new immutable signature. Instead, in the blockchain the deprecation of a resource is impossible due to the immutability feature of blockchain itself, thus the update operation consists in the registration of an ADD transaction for the new version of the resource. The new block associated with the update transaction is linked to the previous one (where the hash of the old version of the resource was stored).
Wallet – DIGITAL	A type of online storage that lets you securely keep track of your digital assets and information, like digital certificates and digital identity information.